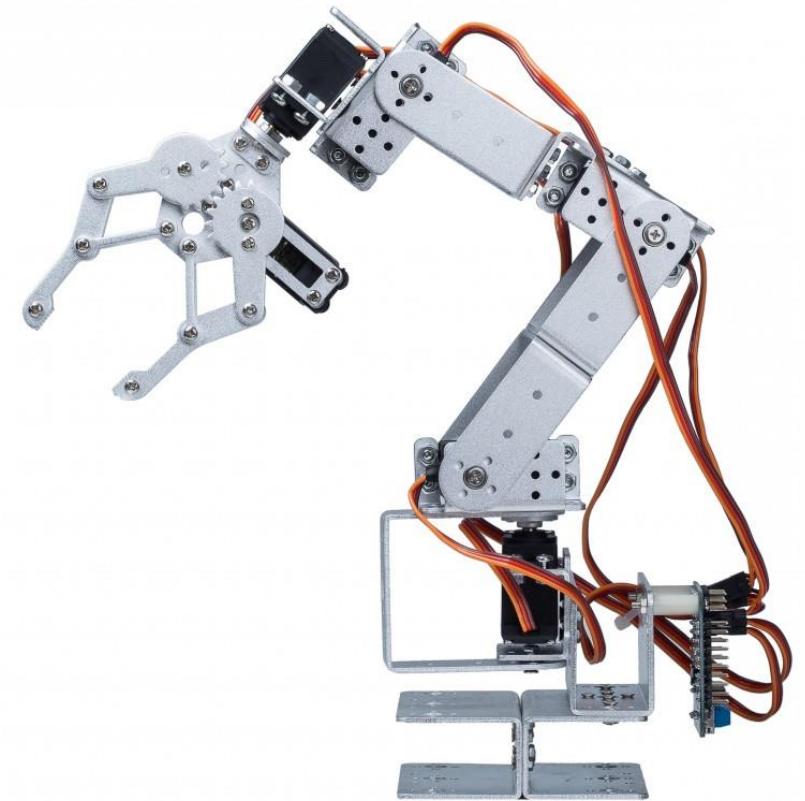




RAI 811: ROBOT MECHANICS & CONTROL

Manipulator

- Manipulator can be thought of as a set of rigid links connected in a chain by joints.
- One end of the chain, the base, is generally fixed and the other end is free to move in space and holds the tool or end-effector that does the useful work.



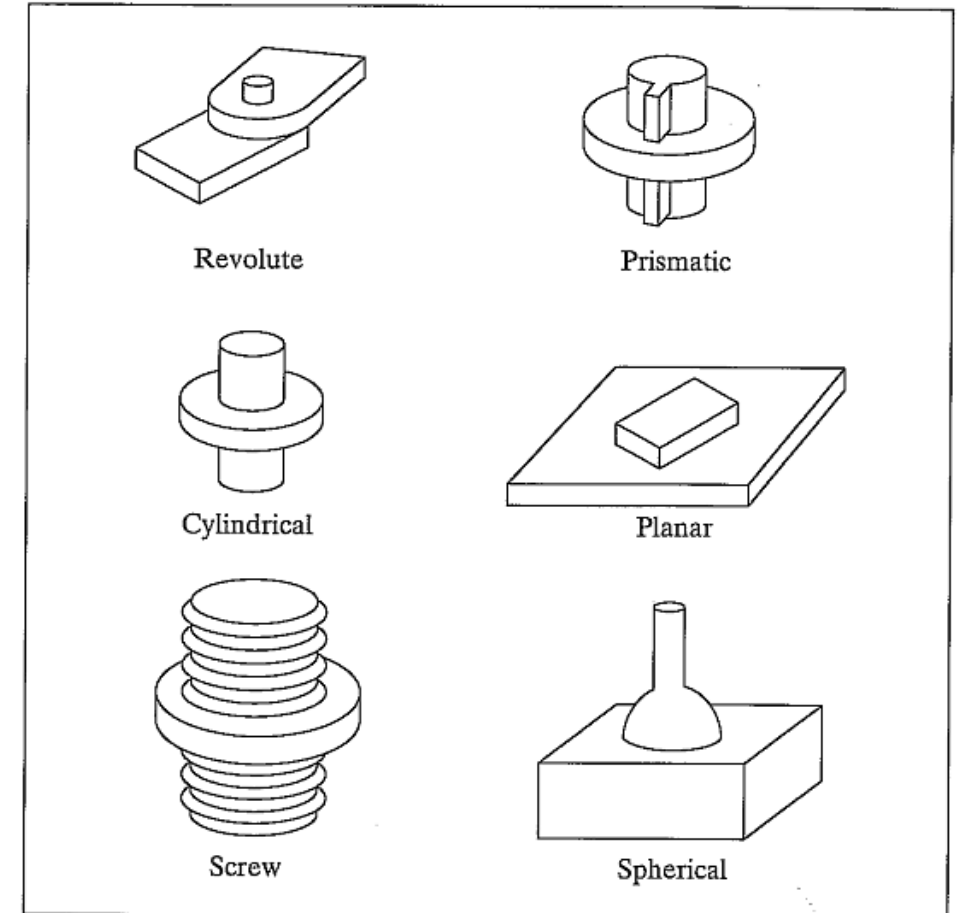
Manipulator



- Types of Joints:

- i. Revolute: 1 DOF (rotation)
- ii. Prismatic: 1 DOF (translation)
- iii. Cylindrical: 2 DOF (1 revolute + 1 prismatic)
- iv. Planar: 3 DOF (1 revolute + 2 prismatic)
- v. Screw: 1 DOF (revolute and prismatic)
- vi. Spherical: 3 DOF (3 revolute)

- DOF = Degrees of Freedom. This is the number of independent movements that a rigid body can have.



Kinematics



- Kinematics is the branch of mechanics that studies the motion of a body or a system of bodies.
- Concerned with position (and angles), velocities (translational and angular) and accelerations (translational and angular).
- Not concerned with force or moments. These are studied in Dynamics.

Manipulator Kinematics

- Two kinematic problems in Robotics.
- **Forward Kinematics:** Given joint angles, we can find the pose of the robot's end-effector.
- **Inverse Kinematics:** Given the pose of robot's end-effector, we can find the corresponding joint angles.

Goals



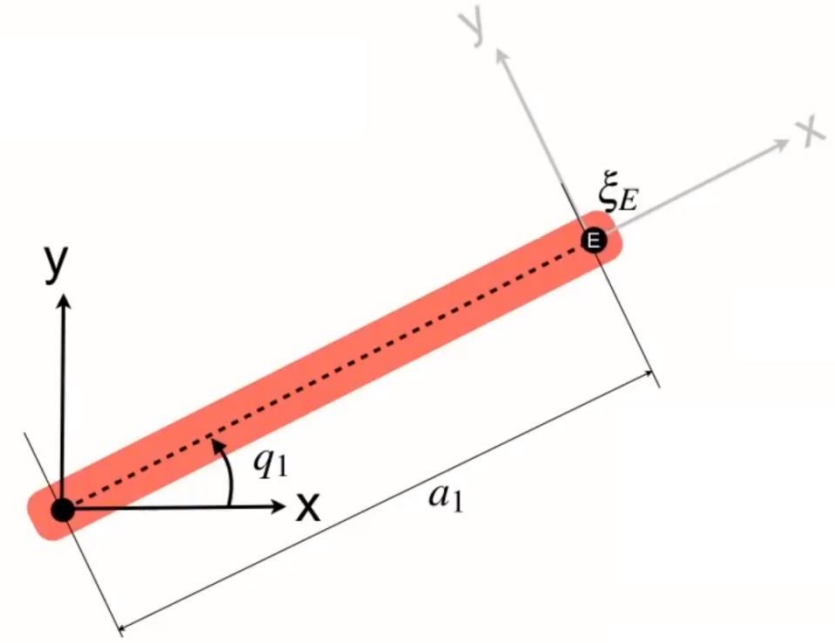
- **Forward Kinematics:** How to compute the pose of the end-effector?
- **Inverse Kinematics:** How to compute the position of each joint given the end-effector pose?
- How to control the velocities and accelerations of the joints in order to generate smooth trajectories for the end-effector?

Link Description

- The links are numbered from the base of the manipulator, called *link 0*
- The remaining links are, in succeeding order, *link 1*, *link 2*, etc.

Forward Kinematics – Single Joint

- Consider the simple robot arm, which has a single rotational joint.
- We can describe the pose of its end-effector – frame {E} – by a sequence of relative poses: a rotation about the joint axis and then a translation by a_1 along the rotated x-axis



$$\mathbf{E} = \mathbf{R}(q_1) \mathbf{T}_x(a_1)$$

$$\mathbf{E} = \begin{pmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cos q_1 & -\sin q_1 & a_1 \cos q_1 \\ \sin q_1 & \cos q_1 & a_1 \sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

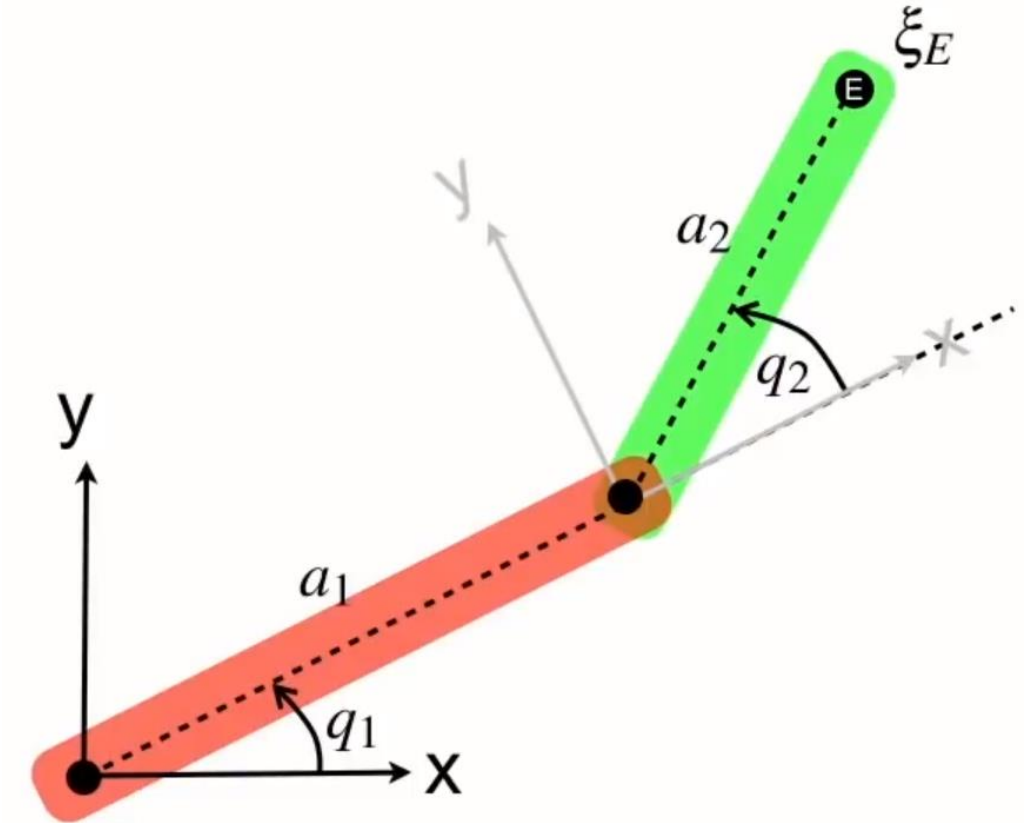
Forward Kinematics – 2 Joints

- The gray frame is a result of $\mathbf{R}(q_1)\mathbf{T}_x(a_1)$

$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1 + q_2) & -\sin(q_1 + q_2) & a_2 \cos(q_1 + q_2) + a_1 \cos q_1 \\ \sin(q_1 + q_2) & \cos(q_1 + q_2) & a_2 \sin(q_1 + q_2) + a_1 \sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

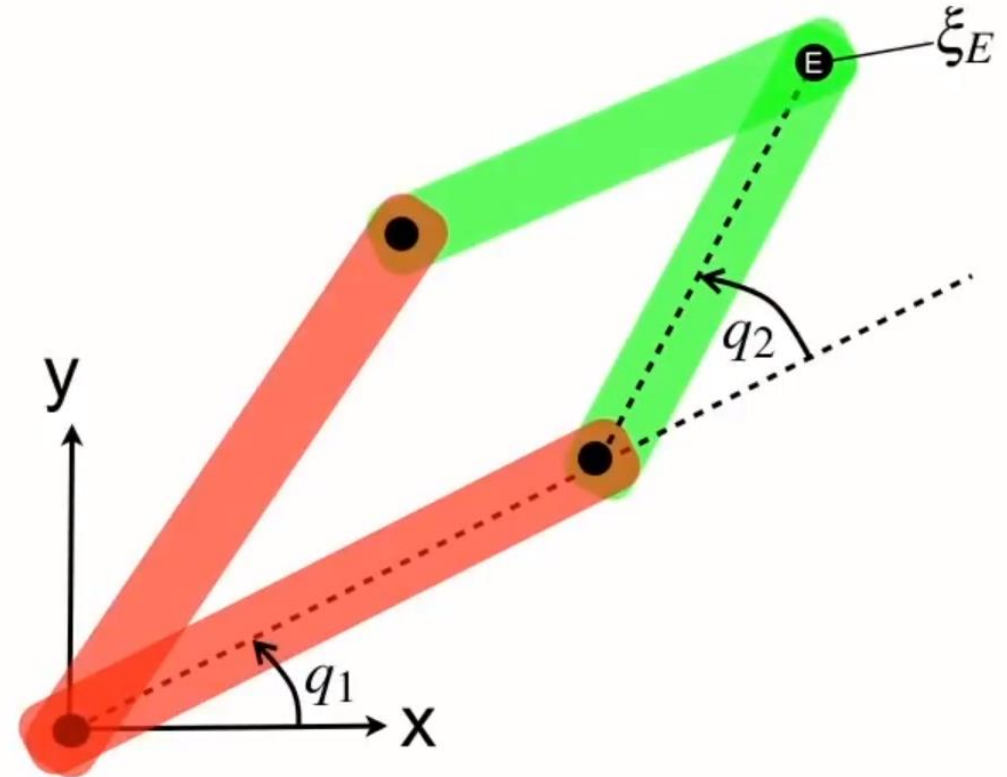
$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_2 \cos(q_1 + q_2) + a_1 \cos q_1 \\ a_2 \sin(q_1 + q_2) + a_1 \sin q_1 \end{pmatrix}$$



Forward Kinematics – 2 Joints



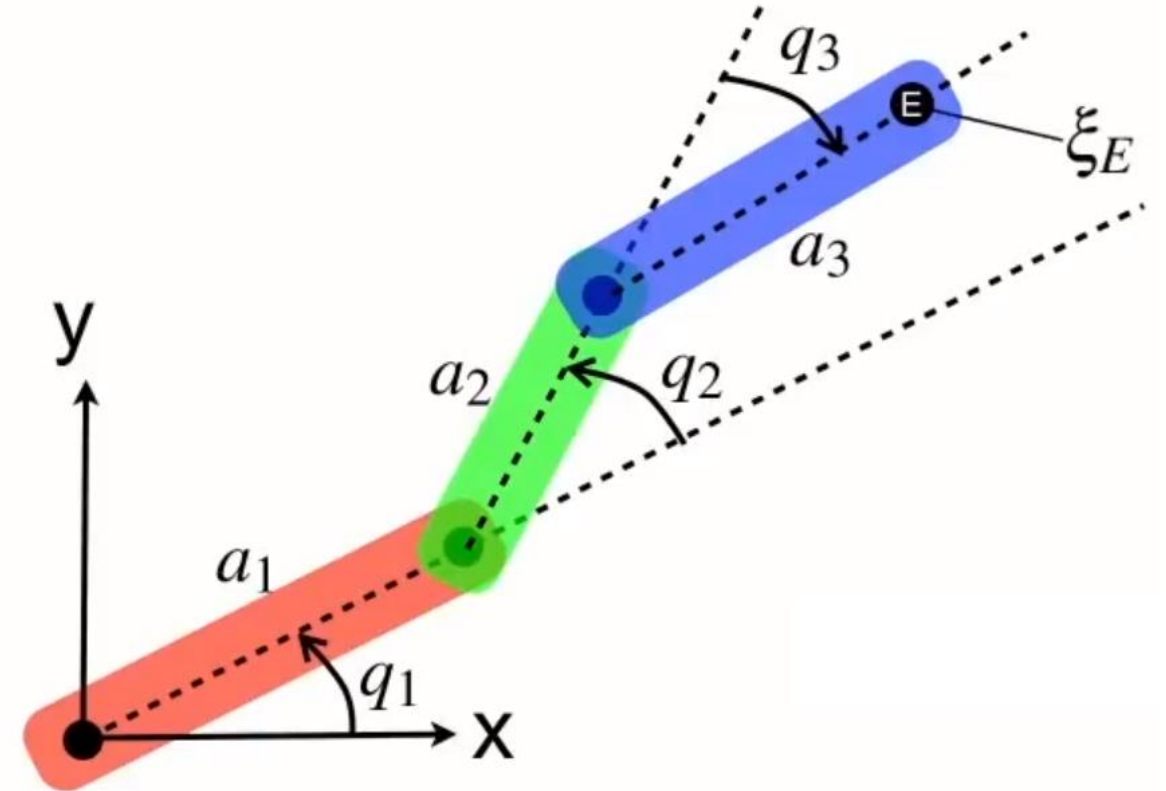
- Configurations are not unique!



Forward Kinematics – 3 Joints



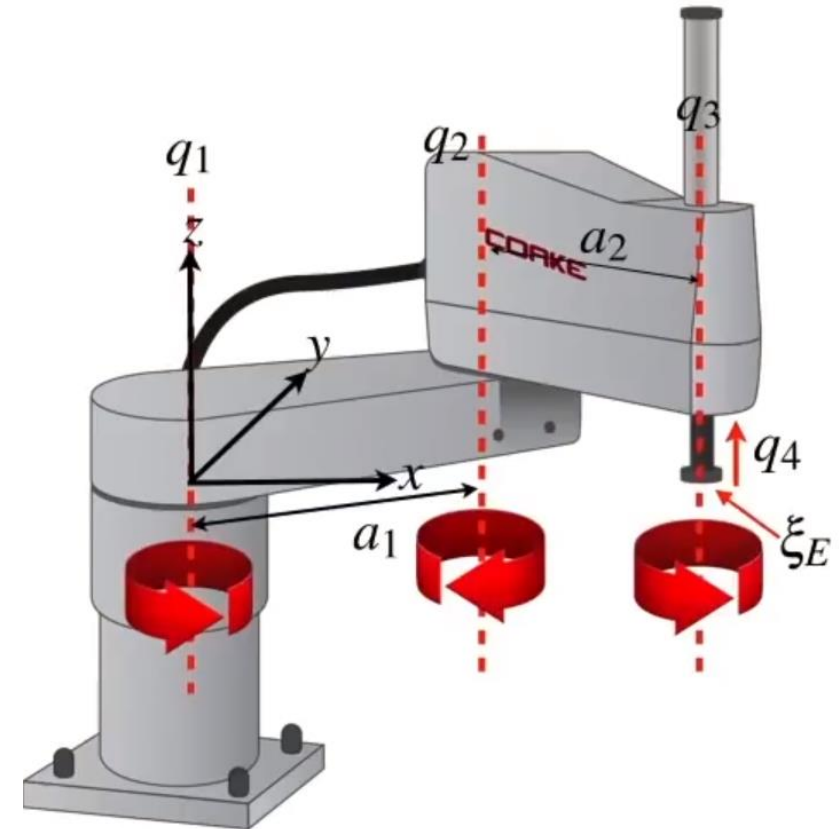
$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\mathbf{T}_x(a_2)\mathbf{R}(q_3)\mathbf{T}_x(a_3)$$



Forward Kinematics – 3D

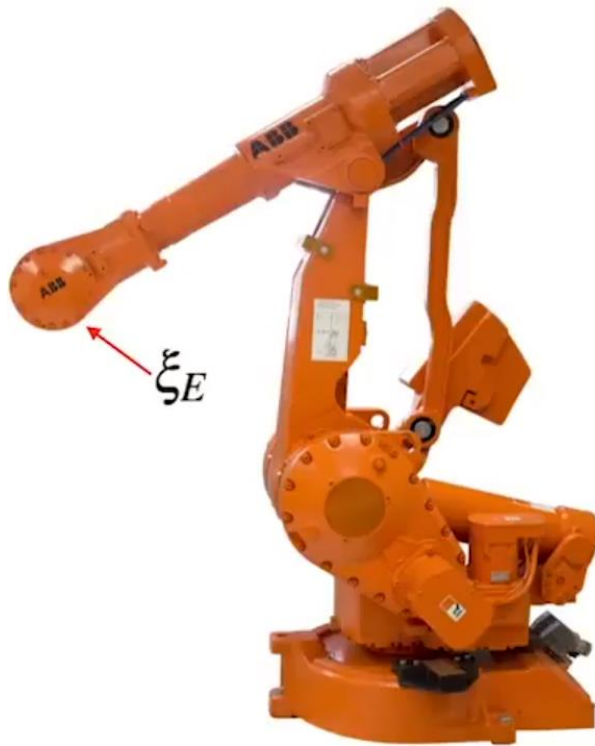
- **SCARA robot.**
- SCARA = Selective Compliance Assembly Robot Arm

$$\mathbf{E} = \mathbf{R}_z(q_1) \mathbf{T}_x(a_1) \mathbf{R}_z(q_2) \mathbf{T}_x(a_2) \mathbf{R}_z(q_3) \mathbf{T}_z(q_4)$$

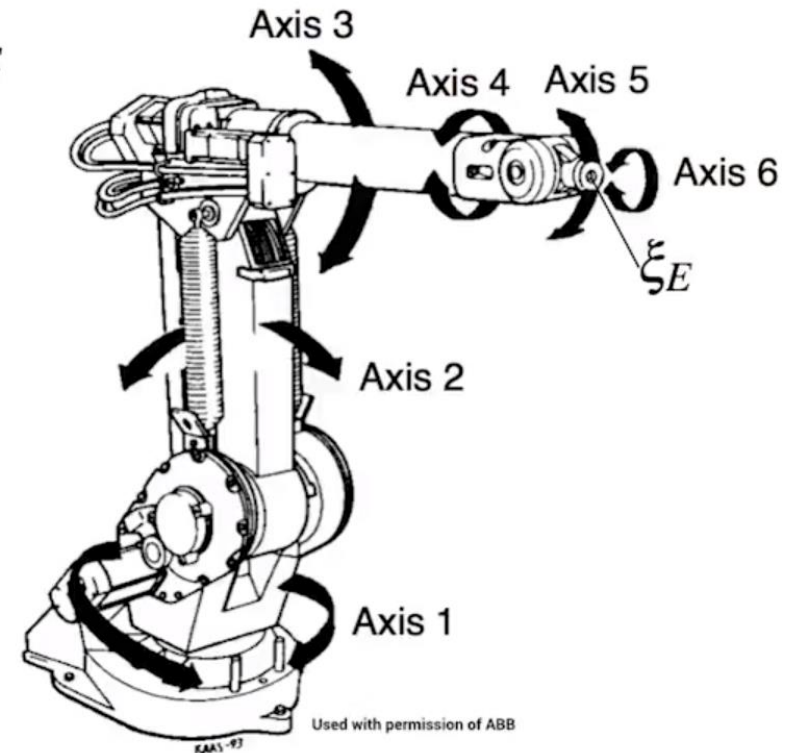


Forward Kinematics – 3D

- Serial Link Manipulator

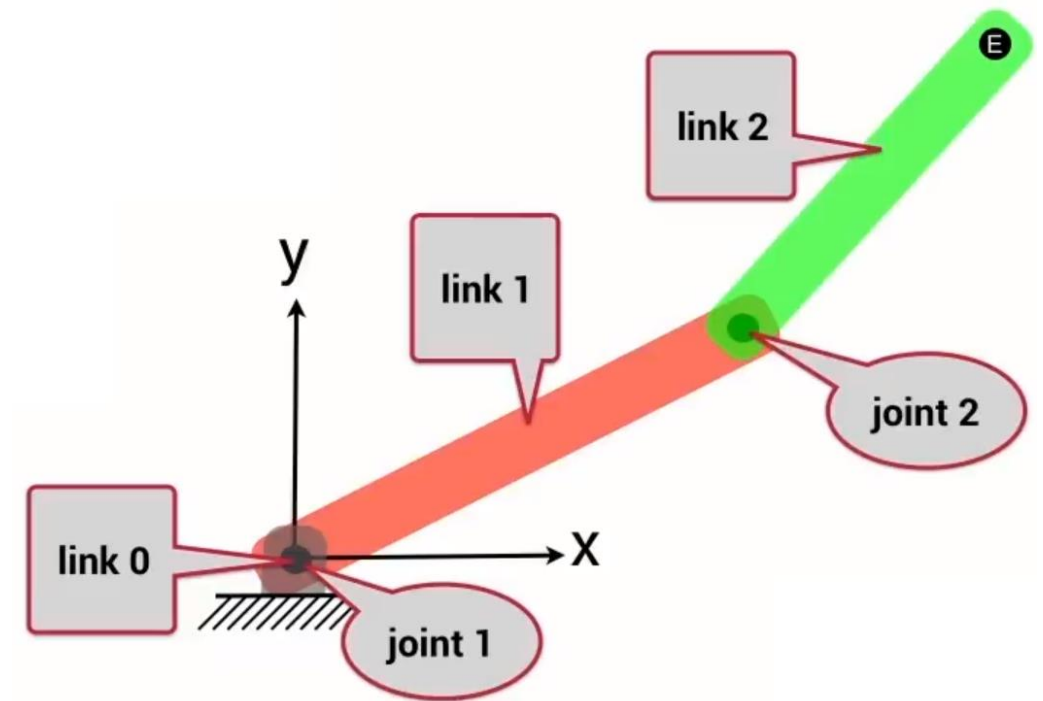


Unimate Puma 500
Qussama Khatib | Used with permission



Serial Link Manipulator

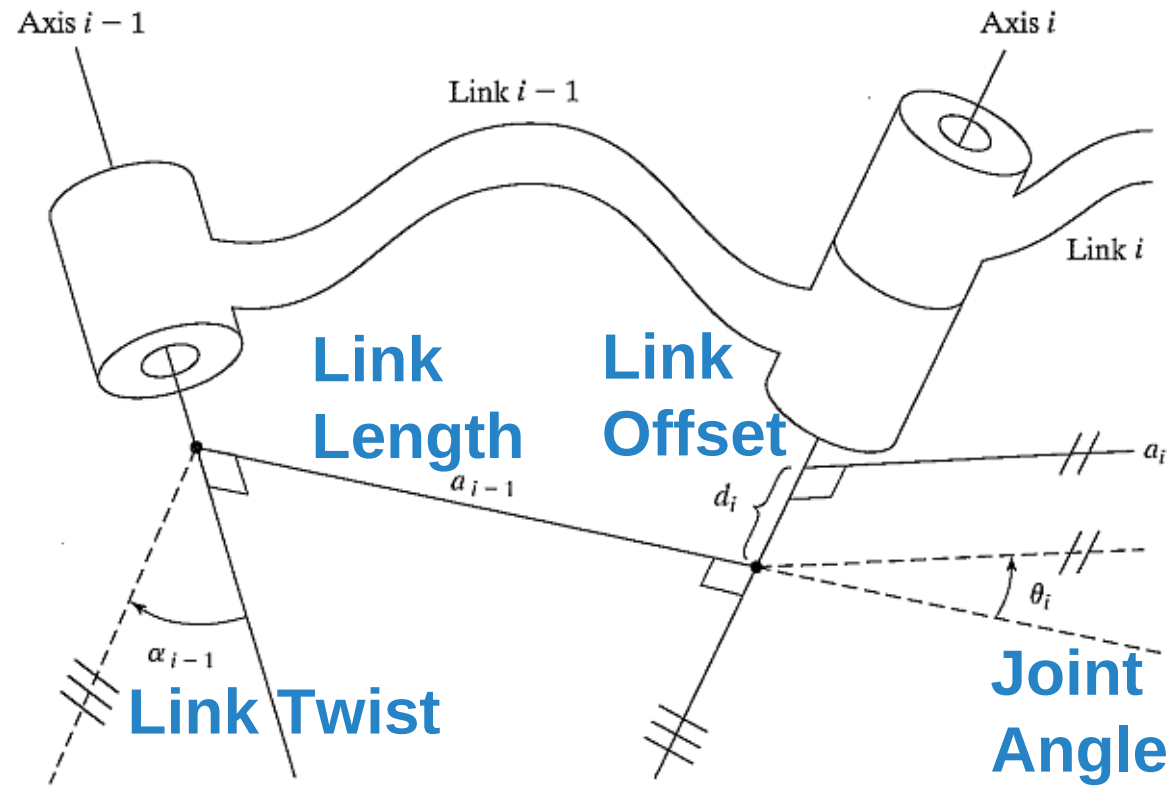
- Each joint attached to another joint via a link
 - Links are rigid
 - Joints can move and are either revolute or prismatic
- Every joint connects two links
- Every link connects two joints
 - excluding first and last link
- A serial manipulator has
 - N joints
 - $N+1$ links



Denavit-Hartenberg Notation

- Developed a general theory to describe an articulated sequence of joints.
- **Goal:** Kinematic modeling of the manipulator(mapping end-effector to base of the robot)
- Each joint in the robot is described by **four parameters**.
- Only applicable to serial link mechanisms. NOT applicable to parallel mechanisms

DH Parameters



DH Parameters

- **Link Length (a):** This distance is measured along a line that is mutually perpendicular to both axes. This mutual perpendicular always exists; it is unique except when both axes are parallel, in which case there are many mutual perpendiculars of equal length.
- **Link Twist (α):** This angle is measured from axis $i-1$ to axis i .

DH Parameters

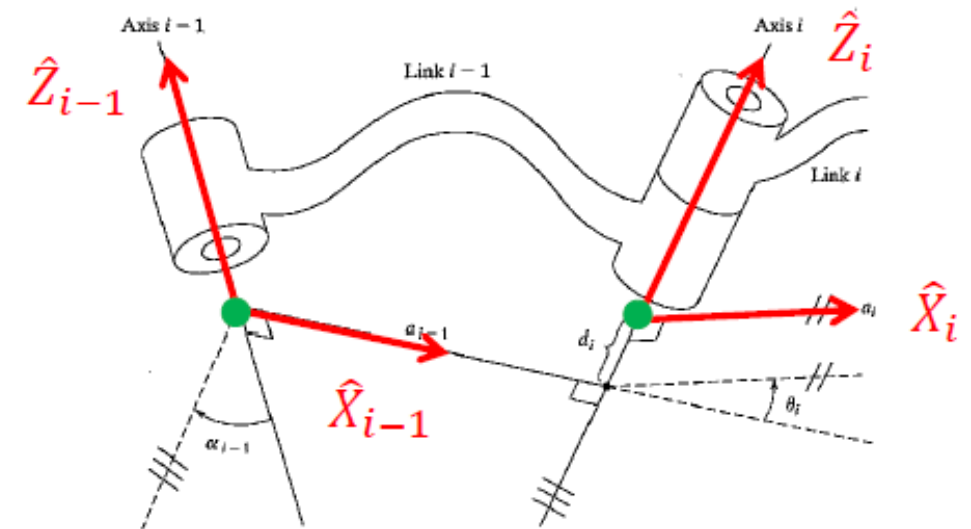
- **Link Offset (d):** Neighboring links have a common joint axis between them. One parameter of interconnection has to do with the distance along this common axis from one link to the next. This parameter is called the link offset.
- **Joint Angle (θ):** parameter that describes the amount of rotation about this common axis between one link and its neighbor. This is called the joint angle

Link Parameters

- The quantities a_i , α_i , d_i , and θ_i are called ***link parameters***
- The kinematics of any robotic manipulator can be described using these four link parameters
- The weight, material, rigidity, shape, and other properties of the mechanism's links are disregarded in kinematics
- The definition of any mechanism using these numbers is called Denavit-Hartenberg (D-H) notation and the numbers are called D-H parameters

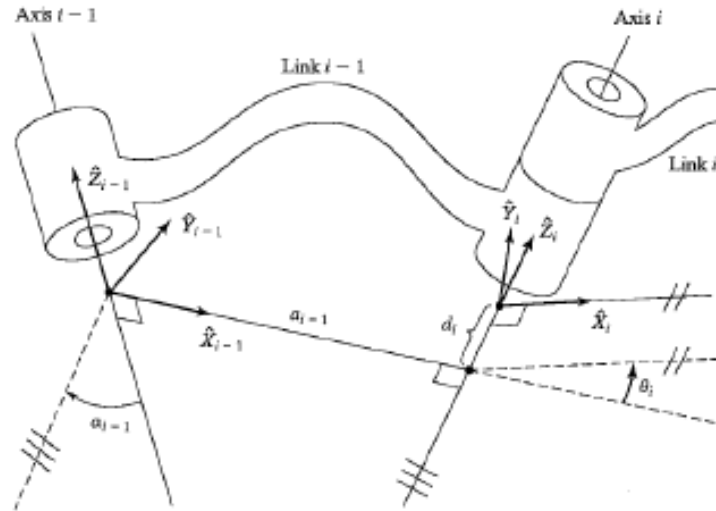
Assigning Frames

- To describe the position and orientation of each link relative to its neighbors, a frame will be assigned to each link
- The frame number will be the same as the link number
- The $\hat{Z}$ -axis of frame $\{i\}$, called $\hat{Z}_i$, is coincident with joint axis i
- The origin of frame $\{i\}$ is the point where a_i intersects joint axis i
- $\hat{X}_i$ points along a_i in the direction from joint i to $i+1$



Assigning Frames – First Frame

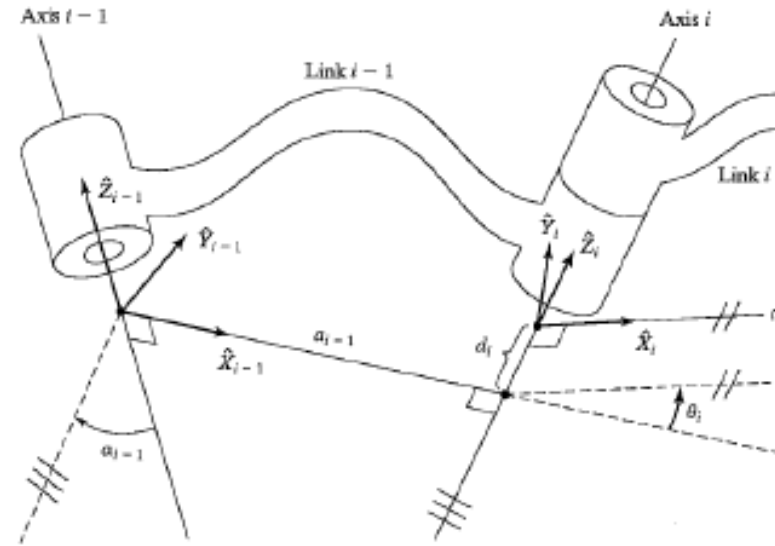
- The immobile first link of the robot is called link 0
- This is assigned the frame {0}
- Frame {0} will be considered the reference frame in all problems
- The position of frame {0} is arbitrary
- For simplicity, the axis $\hat{Z}_0$ is taken along $\hat{Z}_1$, i.e. along the first joint's axis



- Frame {0} is taken as coincident with frame {1} when $\theta_1 = 0$
- Using this convention, $a_0 = 0$ and $\alpha_0 = 0$
- If joint 1 is revolute, $d_1 = 0$
- If joint 1 is prismatic, $\theta_1 = 0$

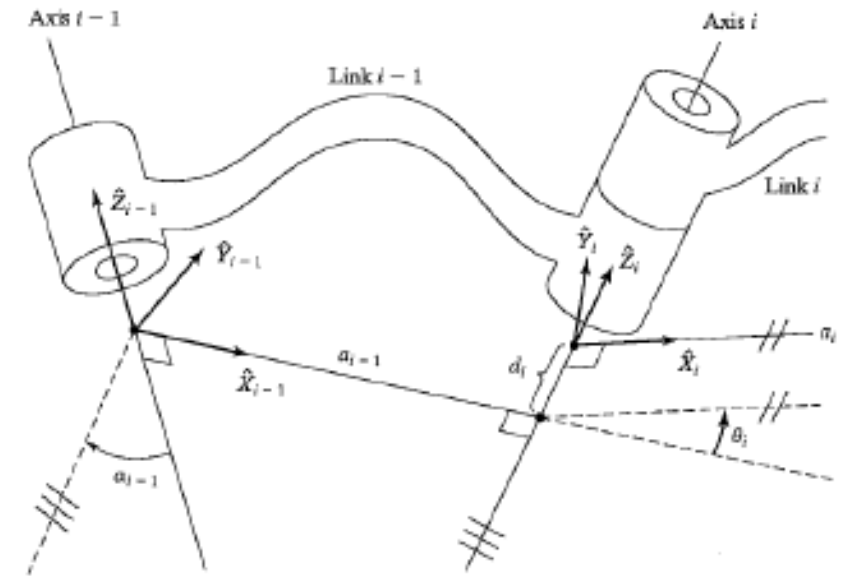
Assigning Frames – Last Frame

- For revolute joint n , $\hat{X}_n$ is chosen so that it aligns with $\hat{X}_{n-1}$ when $\theta_1 = 0$
- The origin of frame $\{N\}$ is chosen so that $d_n = 0$
- If joint n is prismatic, $\hat{X}_n$ is chosen so that $\theta_n = 0$
- The origin of frame $\{N\}$ is chosen at the intersection of $\hat{X}_{n-1}$ and joint axis n when $d_n = 0$



Assigning Frames

- a_i = the distance from $\hat{Z}_i$ to $\hat{Z}_{i+1}$ measured along $\hat{X}_i$
- α_i = the angle from $\hat{Z}_i$ to $\hat{Z}_{i+1}$ about $\hat{X}_i$
- d_i = the distance between $\hat{X}_{i-1}$ and $\hat{X}_i$ measured along $\hat{Z}_i$
- θ_i = the angle from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured about $\hat{Z}_i$



Any Questions?

